HAZBOT / BOUNDED REFERENCE-MODEL VERIFICATION

Abstract command admissibility

01 ENUMERATE

7 phases ×
4,096 Boolean valuations

= 28,672 pairs

per policy, no pruning

02 CONTROLLER

- full guards
- − base–arm interlock
- − target authorization
- − scan age

03 ORACLE

base motion ⇒ S ∧ C ∧ A
approach ⇒ T
arm motion ⇒ B

reach · grip · release

04 VERDICT

- yes → counted
- no → counterexample

shortest prefix by BFS

05 VIOLATING PAIRS PER POLICY

counterexample prefix

All guards	0	—
No base–arm interlock	480	1
No target authorization	128	2
No scan age	288	1

06 REACHABILITY

7/7	phases reachable
13	phase edges incl. self-loops
6	shortest completion depth
11	nominal witness steps → COMPLETE

Violating pairs concern abstract command admissibility, not observed robot failure rates.

Schematic